COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	Charles Vence Espinas	**SECTION:**	CPE31S2
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ACTIVITY NO. 7 **Managing Files and Creating Roles in in Ansible**

1. PROCEDURE OUTPUT

Inventory.yaml

```
 GNU nano 7.2                        inventory.yaml
all:
  vars:
    ansible_user: charles-vence-espinas
  children:

     web_servers:
       hosts:
         server1:
           ansible_host: 192.168.56.108
           os_type: Ubuntu

     db_servers:
       hosts:
         centOS:
           ansible_host: 192.168.56.104
           os_type: CentOS

     file_servers:
       hosts:
         server2:
           ansible_host: 192.168.56.109
           os_type: Ubuntu
```

Task 1: Create a file and copy it to remote servers
1. Using the previous directory was created in your previous laboratory, create a directory, and name it “files.” Create a file inside that directory and name it “default_site.html.” Edit the file and put basic HTML syntax. Any content will do, as long as it will display text later. Save the file and exit.
Example:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>My Web Server</title>
```

</head>
<body>
<h1>Welcome to My Web Server</h1>
This webpage was deployed using Ansible.
This file was copied from the local repository to the remote server.
</body>
</html>

```
  GNU nano 7.2                        default_site.html
<!DOCTYPE HTML>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Web Server</title>
</head>
<body>

  <h1>Welcome to My Web Server</h1>

  <p>This webpage was deployed using Ansible.</p>
  <p>This file was copied from the local repository to the remove server.</p>

</body>
</html>
```

2. Edit the site.yml file and just below the web_servers play, create a new file to copy the default html file for site:
- name: copy default html file for site
tags: apache, apache2, httpd
copy:
src: files/default_site.html
dest: /var/www/html/index.html
owner: root
group: root
mode: '0644'

```
    - name: copy default html file for site
      tags: apache, apache2, httpd
      copy:
        src: files/default_site.html
        dest: /var/www/html/index.html
        owner: root
        group: root
        mode: '0644'
```

3. Run the playbook site.yml. Describe the changes.

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA7$ ansible-playbook -i inventory.yaml site
.yml --ask-become-pass
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [server1]
ok: [centOS]
ok: [server2]

TASK [Install Updates (CentOS)] ************************************************
skipping: [server1]
skipping: [server2]
ok: [centOS]

TASK [Install updates (Ubuntu)] ************************************************
skipping: [centOS]
ok: [server1]
ok: [server2]

PLAY [web_servers] *************************************************************

TASK [Gathering Facts] *********************************************************
ok: [server1]

TASK [Install apache and php for ubuntu servers] *******************************
ok: [server1]

TASK [install apache and php for CentOS servers] *******************************
skipping: [server1]
```

```
TASK [start httpd (CentOS)] ****************************************************
skipping: [server1]

TASK [copy default html file for site] *****************************************
ok: [server1]

PLAY [db_servers] **************************************************************

TASK [Gathering Facts] *********************************************************
ok: [centOS]

TASK [Install mariadb package (CentOS)] ****************************************
ok: [centOS]

TASK [Mariadb- Restarting/Enabling] ********************************************
changed: [centOS]

TASK [install mariadb package (Ubuntu)] ****************************************
skipping: [centOS]

PLAY [file_servers] ************************************************************

TASK [Gathering Facts] *********************************************************
ok: [server2]

TASK [install samba package] ***************************************************
changed: [server2]

PLAY RECAP *********************************************************************
centOS                     : ok=5    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
server1                    : ok=5    changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
server2                    : ok=4    changed=1    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

- **Running the playbook successfully completed the tasks on the remote servers. The webserver was configured with Apaache and PHP, the HTML file was copied, MariaDB was managed on the database server and the samba was installed on the file server.**

4. Go to the remote servers (web_servers) listed in your inventory. Use cat command to check if the index.html is the same as the local repository file (default_site.html). Do both for Ubuntu and CentOS

servers. On the CentOS server, go to the browser and type its IP address. Describe the output.

Webserver hosts(server1)

```
charles-vence-espinas@server1:~$ cat /var/www/html/index.html
<!DOCTYPE HTML>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Web Server</title>
</head>
<body>

  <h1>Welcome to My Web Server</h1>

  <p>This webpage was deployed using Ansible.</p>
  <p>This file was copied from the local repository to the remove server.</p>

</body>
</html>
```



- **Since i assign the server1 in the webserver group, I use that vm which is server 1 to print the output by entering its ip address in the browser and also I display the output also in the terminal by using cat command. So the output in the terminal is the one that we write in default_site.html which is the "Welcome to My Web Browser" and etc and it is an html file format and the output in site is same as the content of the html but rendered as a webpage.**

5. Sync your local repository with GitHub and describe the changes.

- **The updated site.yml and the new default_site.html file were pushed to girhub. This keeps the remote repository updated with the latest changes.**

Task 2: Download a file and extract it to a remote server

1. Edit the site.yml. Just before the web_servers play, create a new play:

- hosts: workstations
become: true
tasks:
- name: install unzip
package:
name: unzip
- name: install terraform
unarchive:
src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28_linux_amd64.zip

dest: /usr/local/bin
remote_src: yes
mode: 0755
owner: root
group: root

```
- hosts: workstations
  become: true
  tasks:
    - name: install unzip
      package:
        name: unzip

    - name: install terrraform
      unarchive:
        src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28_linux_amd64.zip
        dest: /usr/local/bin
        remote_src: yes
        mode: 0755
        owner: root
        group: root
```

note: In Ansible, a play is a section of a playbook that defines which remote machines Ansible should manage and what tasks it should perform on them.
2. Edit the inventory file and add a workstation group. Add any Ubuntu remote server. Make sure to remember the IP address.

```
 GNU nano 7.2                                                inventory.yaml
all:
  vars:
    ansible_user: charles-vence-espinas
  children:

      workstations:
        hosts:
          server1:
            ansible_host: 192.168.56.108
            os_type: Ubuntu
      web_servers:
        hosts:
          server1:
            ansible_host: 192.168.56.108
            os_type: Ubuntu

      db_servers:
        hosts:
          centOS:
            ansible_host: 192.168.56.104
            os_type: CentOS

      file_servers:
        hosts:
          server2:
            ansible_host: 192.168.56.109
            os_type: Ubuntu
```

3. Run the playbook. Describe the output.

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA7$ ansible-playbook -i inventory.yaml site
.yml --ask-become-pass
BECOME password:

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [server1]
ok: [server2]
ok: [centOS]

TASK [Install Updates (CentOS)] ************************************************************************************
skipping: [server1]
skipping: [server2]
ok: [centOS]

TASK [Install updates (Ubuntu)] ************************************************************************************
skipping: [centOS]
ok: [server1]
ok: [server2]

PLAY [workstations] ************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [server1]

TASK [install unzip] ***********************************************************************************************
ok: [server1]
```

```
TASK [install terrraform] ******************************************************************************************
changed: [server1]

PLAY [web_servers] *************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [server1]

TASK [Install apache and php for ubuntu servers] *******************************************************************
ok: [server1]

TASK [install apache and php for CentOS servers] *******************************************************************
skipping: [server1]

TASK [start httpd (CentOS)] ****************************************************************************************
skipping: [server1]

TASK [copy default html file for site] *****************************************************************************
ok: [server1]

PLAY [db_servers] **************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [centOS]

TASK [Install mariadb package (CentOS)] ****************************************************************************
ok: [centOS]
```

```
TASK [Mariadb- Restarting/Enabling] ********************************************************************************
changed: [centOS]

TASK [install mariadb package (Ubuntu)] ****************************************************************************
skipping: [centOS]

PLAY [file_servers] ************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [server2]

TASK [install samba package] ***************************************************************************************
ok: [server2]

PLAY RECAP *********************************************************************************************************
centOS                     : ok=5    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
server1                    : ok=8    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
server2                    : ok=4    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

- **The playbook successfully installed the required unzip package and downloaded and extracted terraform to /usr/local/bin.**

4. On the Ubuntu remote workstation, type terraform to verify installation of terraform. Describe the output.

```
charles-vence-espinas@server1:~$ terraform
Usage: terraform [-version] [-help] <command> [args]

The available commands for execution are listed below.
The most common, useful commands are shown first, followed by
less common or more advanced commands. If you're just getting
started with Terraform, stick with the common commands. For the
other commands, please read the help and docs before usage.

Common commands:
    apply              Builds or changes infrastructure
    console            Interactive console for Terraform interpolations
    destroy            Destroy Terraform-managed infrastructure
    env                Workspace management
    fmt                Rewrites config files to canonical format
    get                Download and install modules for the configuration
    graph              Create a visual graph of Terraform resources
    import             Import existing infrastructure into Terraform
    init               Initialize a Terraform working directory
    login              Obtain and save credentials for a remote host
    logout             Remove locally-stored credentials for a remote host
    output             Read an output from a state file
    plan               Generate and show an execution plan
    providers          Prints a tree of the providers used in the configuration
    refresh            Update local state file against real resources
    show               Inspect Terraform state or plan
    taint              Manually mark a resource for recreation
    untaint            Manually unmark a resource as tainted
    validate           Validates the Terraform files
    version            Prints the Terraform version
    workspace          Workspace management
```

```
All other commands:
    0.12upgrade         Rewrites pre-0.12 module source code for v0.12
    debug               Debug output management (experimental)
    force-unlock        Manually unlock the terraform state
    push                Obsolete command for Terraform Enterprise legacy (v1)
    state               Advanced state management
```

- **The terraform commands displayed its usage information and available commands. This confirmed that the terraform was successfully installed on the ubuntu workstation.**

Task 3: Create roles

1. Edit the site.yml. Configure roles as follows: (make sure to create a copy of the old site.yml file because you will be copying the specific plays for all groups)

```
  GNU nano 7.2                                                site.yml
---
- hosts: all
  become: true
  pre_tasks:

  - name: update repository index (CentOS)
    tags: always
    dnf:
      update_cache: yes
    changed_when: false
    when: ansible_distribution == "CentOS"

  - name: install updates (Ubuntu)
    tags: always
    apt:
      update_cache: yes
    changed_when: false
    when: ansible_distribution == "Ubuntu"

- hosts: all
  become: true
  roles:
    - base

- hosts: workstations
  become: true
  roles:
    - workstations
```

```
- hosts: web_servers
  become: true
  roles:
    - web_servers

- hosts: db_servers
  become: true
  roles:
    - db_servers

- hosts: file_servers
  become: true
  roles:
    - file_servers
```

Save the file and exit.

2. Under the same directory, create a new directory and name it roles. Enter the roles directory and create new directories: base, web_servers, file_servers, db_servers and workstations. For each directory, create a directory and name it tasks.

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA7$ mkdir roles
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA7$ ls
files  inventory.yaml  old_site.yml  roles  rolesOld  site.yml
```

```
vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA7/roles$ mkdir base web_servers file_servers db_
servers workstations
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA7/roles$ ls
base  db_servers  file_servers  web_servers  workstations
```

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA7/roles$ mkdir -p base/tasks web_servers
/tasks file_servers/tasks db_servers/tasks workstations/tasks
```

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA7$ tree roles
roles
├── base
│   └── tasks
├── db_servers
│   └── tasks
├── file_servers
│   └── tasks
├── web_servers
│   └── tasks
└── workstations
    └── tasks

11 directories, 0 files
```

3. Go to tasks for all directory and create a file. Name it main.yml. In each of the tasks for all directories, copy and paste the code from the old site.yml file. Show all contents of main.yml files for all tasks.

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA7/roles$ touch base/tasks/main.yml db_se
rvers/tasks/main.yml file_servers/tasks/main.yml web_servers/tasks/main.yml workstations/tasks/main.yml
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA7/roles$ cd ..
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA7$ tree roles
roles
├── base
│   └── tasks
│       └── main.yml
├── db_servers
│   └── tasks
│       └── main.yml
├── file_servers
│   └── tasks
│       └── main.yml
├── web_servers
│   └── tasks
│       └── main.yml
└── workstations
    └── tasks
        └── main.yml
```

base

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA7$ cat roles/base/tasks/main.yml
---
- name: Install Updates (CentOS)
  tags: always
  dnf:
    update_only: yes
    update_cache: yes
  when: ansible_distribution == "CentOS"

- name: Install updates (Ubuntu)
  tags: always
  apt:
    upgrade: dist
    update_cache: yes
  when: ansible_distribution == "Ubuntu"
```

db_servers

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA7$ cat roles/db_servers/tasks/main.yml
---
- name: Install mariadb package (CentOS)
  tags: centos, db, mariadb
  yum:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "CentOS"

- name: "Mariadb- Restarting/Enabling"
  service:
    name: mariadb
    state: restarted
    enabled: true

- name: install mariadb package (Ubuntu)
  tags: db, mariadb, ubuntu
  apt:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "Ubuntu"
```

file_servers

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA7$ cat roles/file_servers/tasks/main.yml
---
- name: install samba package
  tags: samba
  package:
    name: samba
    state: latest
```

web_servers

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA7$ cat roles/web_servers/tasks/main.yml
---
- name: Install apache and php for ubuntu servers
  tags: apache, apache2, ubuntu
  apt:
    name:
      - apache2
      - libapache2-mod-php
    state: latest
    update_cache: yes
  when: ansible_distribution == "Ubuntu"

- name: install apache and php for CentOS servers
  tags: apache, centos, httpd
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"

- name: start httpd (CentOS)
  tags: apache, centos, httpd
  service:
    name: httpd
    state: started
  when: ansible_distribution == "CentOS"
```

```
- name: copy default html file for site
  tags: apache, apache2, httpd
  copy:
    src: files/default_site.html
    dest: /var/www/html/index.html
    owner: root
    group: root
    mode: '0644'
```

workstations

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA7$ cat roles/workstations/tasks/main.yml
---
- name: install unzip
  package:
  name: unzip

- name: install terrraform
  unarchive:
    src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28_linux_amd64.zip
    dest: /usr/local/bin
    remote_src: yes
    mode: 0755
    owner: root
    group: root
```

4. Run the site.yml playbook and describe the output.

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA7$ ansible-playbook -i inventory.yaml site
.yml --ask-become-pass
BECOME password:

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [server2]
ok: [centOS]
ok: [server1]

TASK [update repository index (CentOS)] ****************************************************************************
skipping: [server1]
skipping: [server2]
ok: [centOS]

TASK [install updates (Ubuntu)] ************************************************************************************
skipping: [centOS]
ok: [server1]
ok: [server2]

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [centOS]
ok: [server2]
ok: [server1]
```

```
TASK [base : Install Updates (CentOS)] ***********************************************************************************
skipping: [server1]
skipping: [server2]
ok: [centOS]

TASK [base : Install updates (Ubuntu)] ***********************************************************************************
skipping: [centOS]
ok: [server1]
ok: [server2]

PLAY [workstations] ******************************************************************************************************

TASK [Gathering Facts] ***************************************************************************************************
ok: [server1]

TASK [workstations : install unzip] **************************************************************************************
ok: [server1]

TASK [workstations : install terrraform] *********************************************************************************
ok: [server1]

PLAY [web_servers] *******************************************************************************************************

TASK [Gathering Facts] ***************************************************************************************************
ok: [server1]

TASK [web_servers : Install apache and php for ubuntu servers] ***********************************************************
ok: [server1]
```

```
TASK [web_servers : install apache and php for CentOS servers] ***********************************************************
skipping: [server1]

TASK [web_servers : start httpd (CentOS)] ********************************************************************************
skipping: [server1]

TASK [web_servers : copy default html file for site] *********************************************************************
ok: [server1]

PLAY [db_servers] ********************************************************************************************************

TASK [Gathering Facts] ***************************************************************************************************
ok: [centOS]

TASK [db_servers : Install mariadb package (CentOS)] *********************************************************************
ok: [centOS]

TASK [db_servers : Mariadb- Restarting/Enabling] *************************************************************************
changed: [centOS]

TASK [db_servers : install mariadb package (Ubuntu)] *********************************************************************
skipping: [centOS]

PLAY [file_servers] ******************************************************************************************************

TASK [Gathering Facts] ***************************************************************************************************
ok: [server2]

TASK [file_servers : install samba package] ******************************************************************************
ok: [server2]
```

```
PLAY RECAP ***************************************************************************************************************
centOS                     : ok=7    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
server1                    : ok=10   changed=0    unreachable=0    failed=0    skipped=4    rescued=0    ignored=0
server2                    : ok=6    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

- **The site.yml playbook was successfully ran all the roles for the different servers. The tasks for the base, workstation, web, database and file servers were completed without error whichshows that tho roles were properly configured.**

COMPUTER ENGINEERING

QUEZON CITY

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What is the importance of creating roles?

- Creating roles is important because it helps to organize ansible tasks into a separate and reusable sections. Because instead of putting all the tasks in a one large playbook, role allows us to group the related tasks together like tasks for web servers, database servers and file servers. This makes the playbook easier to understand, manage and modify. Also, roles can aslo be resuled for different servers or project which saves a lot of time and reduces the repeated work.

2. What is the importance of managing files?

- Managing files is important because system administrators always need to create, copy, modify, delete and set permissions for files on different servers. Ansible mames these tasks easier because wr can manage files on multiple remote servers from one control node. This helps keep the servers consistent and reduces the need to manually log in to each server. It makes the server configuration faster, organized and repeatable.

Conclusions/Learnings:

In this laboratory activity, I learned how to use the ansible to manage the files and organizes tasks using roles. I also learned how to copy the files to remote servers and download and extract the files like terraform using an ansible. I also learned how to create roles and organize tasks for different types of servers like web servers, database servers. File servers and workstations. Using roles makes the playbook more organized, reusable and easier to manage. Overall. This activity helped me to understand how an ansible can make the server management faster, more consistent and eraser by reducing the need to perform tasks manually on each server.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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